Lovable Dashboard

Link: https://executive-analytics-suite.lovable.app

1. Project Overview

This project analyzes historical sales data to evaluate product performance, pricing strategies, profitability, and customer demand. Using SQL in Databricks, I developed key business metrics and analytical models to identify sales trends, promotional periods, and price elasticity of demand. Thereafter, I used Lovable to create an interactive dashboard to visualise the business metrics and business objectives below.

2. Business Objectives

The analysis was conducted to answer the following business questions:

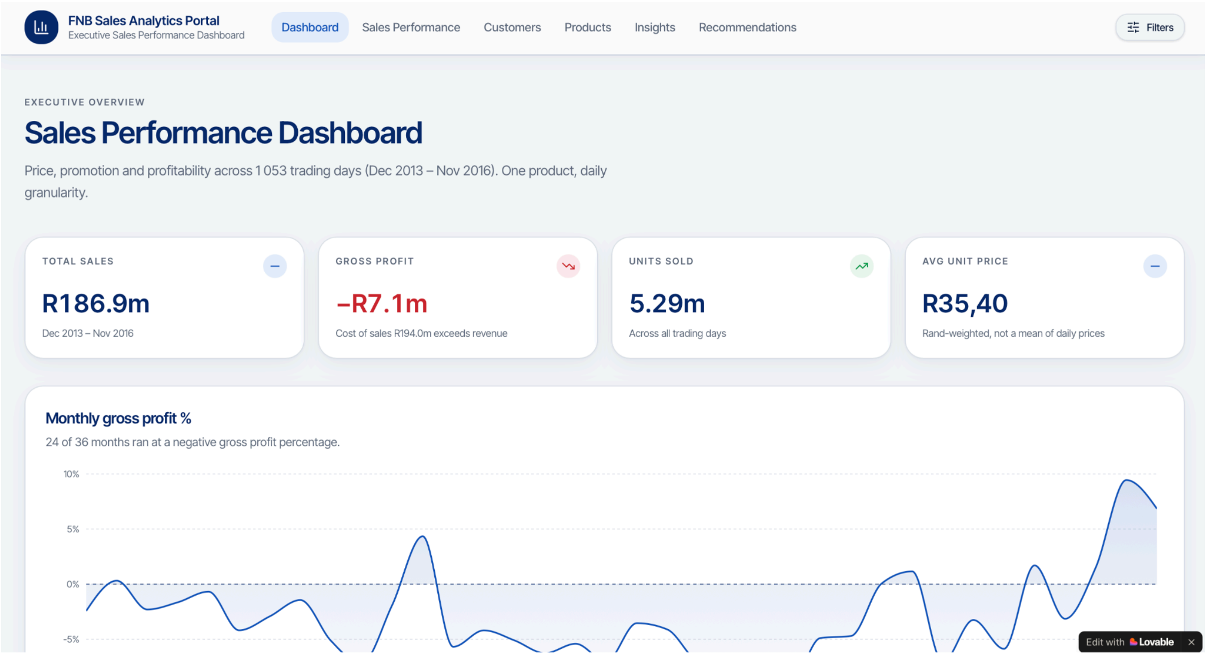
- What is the daily sales price per unit?
- What is the average unit sales price?
- What is the daily gross profit percentage?
- What is the daily gross profit per unit?
- Which promotional periods can be identified from the data?
- What is the Price Elasticity of Demand during these promotional periods?
- Does the product perform better when sold at a promotional price?

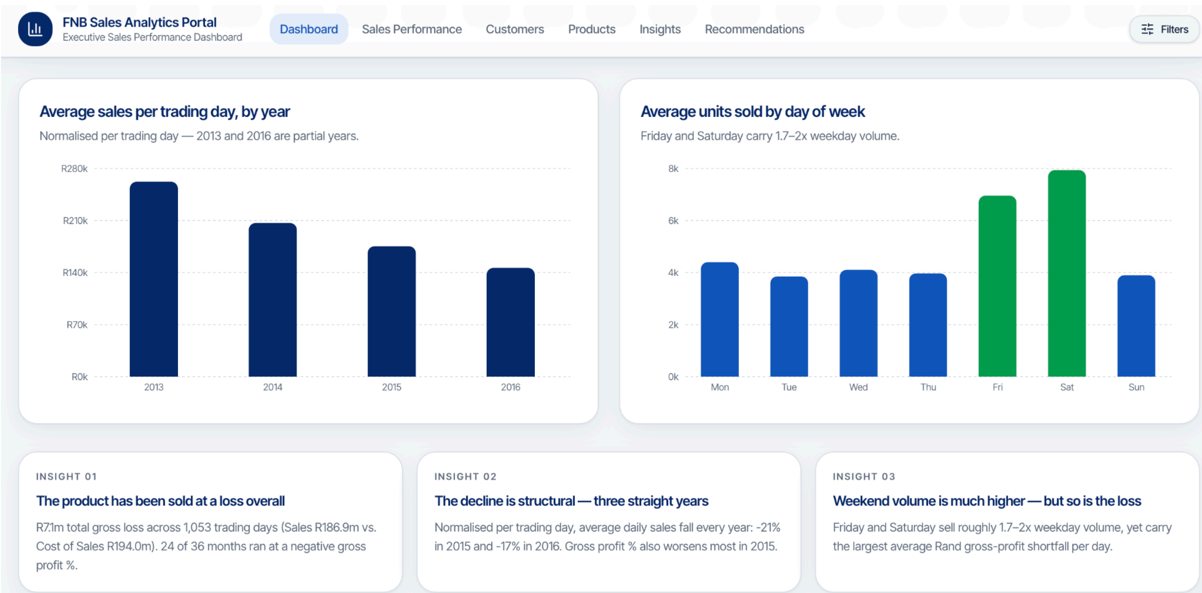
3. Dataset Overview

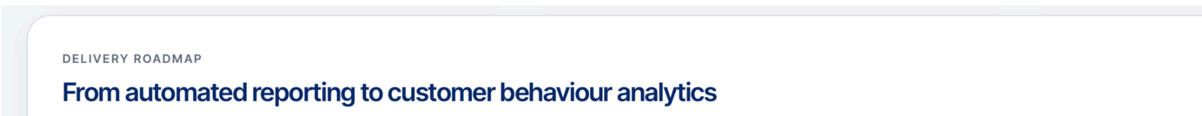
Dataset Information

- Source
- Number of records
- Date range
- Key fields

4.Screenshots of Dashboards







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RECOMMENDATIONS

Where to act first

Prioritised actions arising from the price, promotion and profitability analysis.

01 — Critical

Pause further promotions on this product

The base business is loss-making. More promotions on a product that is already underwater accelerate the loss rather than fix it.

02 — High

Investigate cost-of-sales drivers first

Supplier pricing, shrinkage and cost allocation should be reviewed before any new discounting cycle. Fix margin, then promotions become a genuinely good lever.

03 — High

Prioritise the margin fix on weekend trading

Friday and Saturday absorb the biggest Rand losses. Margin remediation matters most on the highest-volume days, not less.

04 — Medium

Instrument promotions with an explicit promo flag

Promo windows had to be inferred from price alone via gaps-and-islands detection. Capturing a promo indicator at source removes that ambiguity.

DELIVERY ROADMAP

From automated reporting to customer behaviour analytics

A four-phase build sequence for the FNB sales analytics capability.

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INSIGHTS

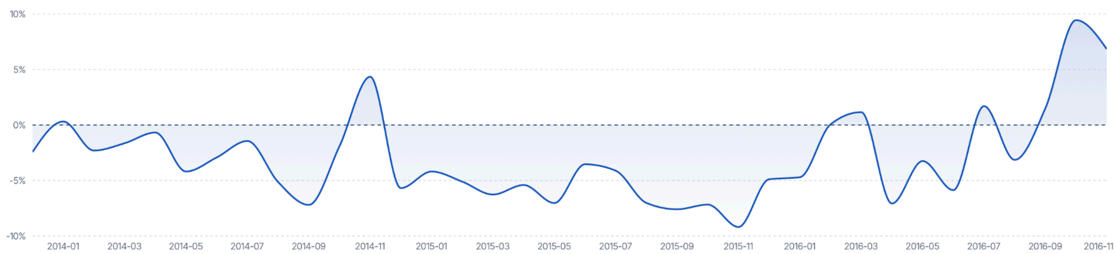
What the data actually says

Three findings beyond the original brief — including the two that changed the executive conclusion.

The product has been sold at a loss overall

Insight 01

R7.1m total gross loss across 1,053 trading days (Sales R186.9m vs. Cost of Sales R194.0m). 24 of 36 months ran at a negative gross profit %.



PRODUCTS

Core product unit economics

One product, three years of daily trading. Revenue holds scale, but cost of sales exceeds it across the full period.

REVENUE	COST OF SALES	GROSS MARGIN	AVG COST PER UNIT
R186.9m	R194.0m	-3.80%	R36,67
Total sales	Exceeds revenue by R7.1m	Negative across the full period	vs. R35,40 avg price

Product ledger

Aggregated across 1,053 trading days.

Product	Units	Revenue	Cost of sales	Gross profit	GP %
Core Product	5 290 000	R186.9m	R194.0m	−R7.1m	-3.80%

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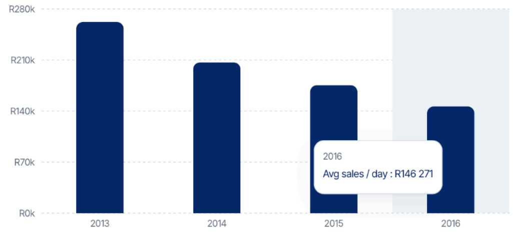
Core Product 5 290 000 R186.9m R194.0m −R7.1m -3.80%

Monthly gross profit %

24 of 36 months negative.

Average sales per trading day, by year

Declining every year.



Only one product is present in the source extract, so category-level comparison is unavailable. The Product Category filter is wired for the multi-product data model arriving with Phase 2.

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CUSTOMERS

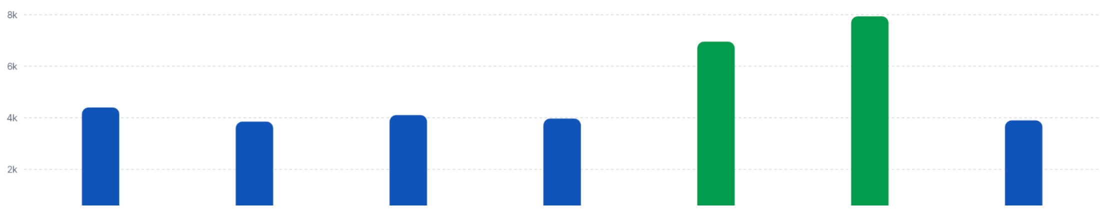
Demand behaviour

The current extract is transactional and un-identified, so customer behaviour is observed through purchase timing and promotional responsiveness rather than through segments.

AVG WEEKEND UNITS / DAY	AVG WEEKDAY UNITS / DAY	PEAK TRADING DAY	PROMO RESPONSIVENESS
7 446	4 045	Saturday	Very high
Friday and Saturday combined average	Monday–Thursday and Sunday	7,937 average units sold	PED between −11 and −46

Average units sold by day of week

Weekend peaks are highlighted — these days also carry the largest Rand gross-profit shortfall.



Highly price-sensitive demand

Discounts of 6–15% produced volume increases of 65–504%. Buying behaviour is dominated by price rather than loyalty in this dataset.

Concentrated weekend purchasing

Friday and Saturday account for the largest share of units. Any service, stock or margin intervention should be weighted to those days.

No identity-level view yet

Retention, basket and segment analytics require customer keys, which arrive with Phase 4 of the roadmap.

Day-of-week detail

MON	TUE	WED	THU	FRI	SAT	SUN
4 400	3 852	4 106	3 969	6 955	7 937	3 897
avg units	avg units	avg units	avg units	avg units	avg units	avg units

Customer segment, province and channel breakdowns are not present in the source extract. Those views become populated once customer-level keys are onboarded in Phase 4 — no segment-level figures are estimated here.

SALES PERFORMANCE

Price, promotion and elasticity

Promotional windows were inferred from price alone using a one-standard-deviation flag, validated against a 14-day trailing rolling average and grouped with gaps-and-islands logic.

PROMOTIONAL PERIODS	STRONGEST ELASTICITY	PEAK VOLUME LIFT	REVENUE AT RISK
3	−46.2	+503.9%	R7.1m
2 of 3 independently confirmed	PED, Sep–Oct 2014	Sep–Oct 2015	Cumulative gross loss to date

Price elasticity of demand

Each promotional period compared against its own 14-day baseline.

Period	Dates	Δ Price	Δ Quantity	PED	Validation
Period 1	24 Sep – 6 Oct 2014	-7.4%	+340.7%	-46.2	Confirmed
Period 2	23 Sep – 6 Oct 2015	-14.7%	+503.9%	-34.4	Confirmed
Period 3	23 Jun – 6 Jul 2016	-5.7%	+64.8%	-11.3	Manual review

All three periods show |PED| far above 1 — demand is highly elastic. A small percentage price cut drove a much larger percentage increase in units sold. Elasticity this extreme is unusual for real-world retail; treat it as a feature of this dataset.

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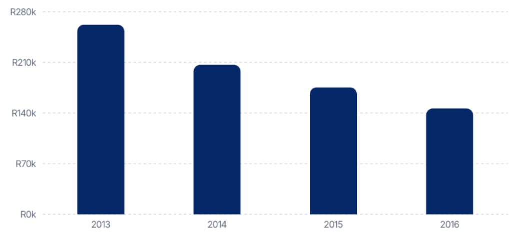
Monthly gross profit %

Margin compression through 2015, recovering late 2016.

Average sales per trading day

−21% in 2015, −17% in 2016.



Methodology — daily unit economics

Metric	Formula
Price per unit	Sales ÷ Quantity Sold
Average unit sales price	SUM(Sales) ÷ SUM(Quantity) — Rand-weighted
Daily % gross profit	(Sales − Cost of Sales) ÷ Sales
% gross profit per unit	(Gross Profit ÷ Qty) ÷ (Sales ÷ Qty)